FERMAT'S EQUATION IN NON-COMMUTATIVE DIVISION ALGEBRAS

TIM JACOBY

Abstract. Fermat's Last Theorem asserts that $a^p + b^p = c^p$ has no nontrivial integer solutions for $p \geq 3$. We prove that this equation *does* have nontrivial solutions in the integer quaternions $\mathbb{H}_\mathbb{Z}$ and integer octonions $\mathbb{O}_\mathbb{Z}$ for every odd prime p. Three independent mechanisms break the theorem: anti-commutativity of imaginary units (odd exponents, dimension ≥ 4), mutual zero divisors (all exponents, dimension ≥ 16), and conjugate-pair cancellation in certain commutative extensions. We establish a sharp dimension threshold: among normed division algebras over $\mathbb{R}$, Fermat's equation is solvable if and only if the dimension is at least 4. All main theorems are machine-verified in Lean 4 with Mathlib: 28 files, 8,791 lines, 355 theorems, zero `sorry`, zero axioms.

1. Introduction

Fermat's Last Theorem (FLT), proved by Wiles [27] in 1995, states that for integers $n \geq 3$, the equation $a^n + b^n = c^n$ has no solutions with $a, b, c \in \mathbb{Z} \setminus \{0\}$. The proof marshals the full force of the modularity theorem for semistable elliptic curves over $\mathbb{Q}$, Ribet's level-lowering theorem, and the dimension-zero space $S_2(\Gamma_0(2))$. Every step depends, explicitly or implicitly, on the commutativity of $\mathbb{Z}$.

This paper asks: what happens when commutativity is removed?

The integer quaternions $\mathbb{H}_\mathbb{Z}$ and integer octonions $\mathbb{O}_\mathbb{Z}$ are the natural testing ground. These are the integer subrings of the only normed division algebras over $\mathbb{R}$ beyond $\mathbb{R}$ and $\mathbb{C}$ (Hurwitz [15]). The quaternions (dimension 4) are associative but non-commutative; the octonions (dimension 8) are neither commutative nor associative, though they are alternative (every two elements generate an associative subalgebra, by Artin's theorem).

The main result is:

Theorem 1.1 (Main Theorem)**.** *For every odd integer $p \geq 3$, the equation $a^p + b^p = c^p$ has nontrivial solutions in both $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$. Specifically, if u, v are elements of any ring R satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$, then*

$$(1 + u)^p + (-1 + v)^p = \varepsilon \cdot (u + v)^p$$

Date: March 2026.

2020 *Mathematics Subject Classification.* Primary 11D41; Secondary 11R52, 17A75, 68V15, 11F70.

Key words and phrases. Fermat's Last Theorem, quaternion integers, octonion integers, non-commutative algebra, formal verification, Lean 4, Langlands program, G_2.

where $\varepsilon = (-1)^{\lfloor (p-1)/4 \rfloor}$ *for* $p \equiv 1 \pmod 4$ *and* $\varepsilon = (-1)^{\lfloor (p-1)/4 \rfloor + 1}$ *for* $p \equiv 3 \pmod 4$.

The proof is purely algebraic: it requires only distributivity, the relations $u^2 = v^2 = -1$, $uv + vu = 0$, and an induction on the exponent. The quaternion integers $\mathbb{H}_\mathbb{Z}$ and Cayley integers $\mathbb{O}_\mathbb{Z}$ both contain such pairs — any two distinct imaginary basis units suffice.

Three mechanisms break FLT outside $\mathbb{Z}$:

(A) **Anti-commutativity.** For $u^2 = v^2 = -1$ with $uv + vu = 0$, the real parts of $(1+u)^p$ and $(-1+v)^p$ cancel exactly. This requires dimension ≥ 4 (two independent imaginary directions) and works for all odd p.

(B) **Zero divisors.** If $ab = ba = 0$ with $a, b \neq 0$, then $(a+b)^n = a^n + b^n$ for all $n \geq 1$. This requires dimension ≥ 16 (the sedenions are the first Cayley-Dickson algebra with zero divisors) and works for *all* exponents, including even ones.

(C) **Conjugate pairs.** In certain commutative extensions of $\mathbb{Z}$ (e.g., $\mathbb{Z}[\sqrt{-5}]$), conjugate-pair cancellation produces cube-sum identities. This mechanism is commutative and does not require non-commutativity.

The dimension threshold is sharp:

Theorem 1.2 (Dimension Threshold)**.** *Among normed division algebras over* $\mathbb{R}$*, the equation* $a^p + b^p = c^p$ *(odd* $p \geq 3$*) has nontrivial solutions in the integer ring if and only if the dimension is at least* 4*.*

The forward direction is Theorem 1.1. The reverse direction combines Wiles' theorem ($\mathbb{R}$, dimension 1) with FLT over $\mathbb{Z}[i]$ ($\mathbb{C}$, dimension 2), which follows from Ward [26] and the single imaginary direction in $\mathbb{C}$ that precludes anti-commuting pairs.

Prior work on Fermat's equation outside $\mathbb{Z}$ includes: Ribenboim [24] on Fermat over p-adic quaternion fields, Ward [26] on the non-existence of non-trivial Gaussian integer cube sums, Li–Yuan [20] on Fermat over 2×2 integer matrices, Pollack [22] on Waring-type results for quaternion cubes, and Conway–Smith [7] on quaternion and octonion integer arithmetic. The all-odd-p symbolic proof (Theorem 1.1), the dimension threshold (Theorem 1.2), and the sedenion phase transition (Theorem 7.1) appear to be new.

All main theorems are verified in Lean 4 with the Mathlib library. The formalization comprises 28 files, 8,791 lines, and 355 theorems with zero `sorry` and zero axioms.

2. Preliminaries

2.1. **Normed division algebras.** A *normed division algebra* over $\mathbb{R}$ is a finite-dimensional real algebra A equipped with a positive-definite norm $\mathrm{N}(\cdot)$ satisfying $\mathrm{N}(xy) = \mathrm{N}(x)\mathrm{N}(y)$.

Theorem 2.1 (Hurwitz, 1898)**.** *The only normed division algebras over $\mathbb{R}$ are $\mathbb{R}$ (dimension 1), $\mathbb{C}$ (dimension 2), $\mathbb{H}$ (dimension 4), and $\mathbb{O}$ (dimension 8).*

These four algebras sit in the Cayley-Dickson tower: each is obtained from the previous by a doubling construction $A \mapsto A \oplus A\ell$ with multiplication $(a, b)(c, d) = (ac - \bar{d}b,\, da + b\bar{c})$. At each step, a structural property is lost: $\mathbb{R} \to \mathbb{C}$ loses ordering, $\mathbb{C} \to \mathbb{H}$ loses commutativity, $\mathbb{H} \to \mathbb{O}$ loses associativity. The octonions retain *alternativity*: every subalgebra generated by two elements is associative (Artin's theorem).

2.2. **Integer subrings.** The *Lipschitz integers* $\mathbb{H}_\mathbb{Z} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{Z}\}$ form a subring of $\mathbb{H}$ closed under multiplication. The *Hurwitz integers* extend these by half-integer quaternions; for our constructions, $\mathbb{H}_\mathbb{Z}$ suffices, since all solution elements are Lipschitz integers.

The *Cayley integers* $\mathbb{O}_\mathbb{Z}$ are the octonionic analogue. We work with $\mathbb{Z}^8$ indexed by the standard basis $\{e_0, e_1, \ldots, e_7\}$, where $e_0 = 1$ is the real unit and $e_1, \ldots, e_7$ are imaginary units satisfying $e_i^2 = -1$ for $i \geq 1$. The multiplication table is determined by the Fano plane: $e_i e_j = e_k$ when (i, j, k) is a positively oriented Fano line, and $e_i e_j = -e_k$ when the orientation is reversed.

2.3. **Key identities.** The following identities hold in any ring R with elements u, v satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$.

Lemma 2.2. $(1 + u)^2 = 2u$ *and* $(-1 + v)^2 = -2v$.

Proof. $(1 + u)^2 = 1 + u + u + u^2 = 1 + 2u - 1 = 2u$. Similarly, $(-1 + v)^2 = 1 - v - v + v^2 = 1 - 2v - 1 = -2v$. □

Lemma 2.3. $(u + v)^2 = -2$.

Proof. $(u + v)^2 = u^2 + uv + vu + v^2 = -1 + 0 + (-1) = -2$. □

These identities imply that powers of $1 + u$, $-1 + v$, and $u + v$ grow in a controlled way, enabling the induction in Section 4.

3. The cubic theorem

Theorem 3.1 (Cubic Fermat in $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$)**.** *Let $a = 1 + e_1$, $b = -1 + e_2$, $c = -e_1 - e_2$. Then $a^3 + b^3 = c^3$ in both $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$. All three elements are nonzero with $\mathrm{N}(a) = \mathrm{N}(b) = \mathrm{N}(c) = 2$.*

Proof. In the subalgebra $\mathbb{R} \oplus \mathbb{R}e_1 \cong \mathbb{C}$, write $a = 1 + e_1$. Since $e_1^2 = -1$:

$$a^3 = (1 + e_1)^3 = 1 + 3e_1 + 3e_1^2 + e_1^3 = 1 + 3e_1 - 3 - e_1 = -2 + 2e_1.$$

Similarly, in $\mathbb{R} \oplus \mathbb{R}e_2$, write $b = -1 + e_2$:

$$b^3 = (-1 + e_2)^3 = -1 + 3e_2 - 3e_2^2 + e_2^3 = -1 + 3e_2 + 3 - e_2 = 2 + 2e_2.$$

Therefore $a^3 + b^3 = (-2 + 2e_1) + (2 + 2e_2) = 2e_1 + 2e_2$.

For $c = -e_1 - e_2$, we use Lemma 2.3: $(e_1 + e_2)^2 = -2$, so

$$c^2 = (e_1 + e_2)^2 = -2, \qquad c^3 = c^2 \cdot c = (-2)(-e_1 - e_2) = 2e_1 + 2e_2.$$

Hence $a^3 + b^3 = c^3$.

The norms are $\mathrm{N}(a) = 1^2+1^2 = 2$, $\mathrm{N}(b) = 1^2+1^2 = 2$, $\mathrm{N}(c) = 1^2+1^2 = 2$. All three elements are nonzero. □

Remark 3.2. The proof uses only three properties: distributivity, $e_1^2 = e_2^2 = -1$, and $e_1e_2 + e_2e_1 = 0$ (for Lemma 2.3). It does not use associativity of the ambient algebra. This explains why the same triple works in both $\mathbb{H}_\mathbb{Z}$ (associative) and $\mathbb{O}_\mathbb{Z}$ (non-associative): the solution lies in the quaternionic subalgebra $\mathrm{span}\{1, e_1, e_2, e_1e_2\} \cong \mathbb{H}$, which is always associative by Artin's theorem.

Remark 3.3. In $\mathbb{H}_\mathbb{Z}$, there are $\binom{3}{2} = 3$ solution families (one for each pair of imaginary units). In $\mathbb{O}_\mathbb{Z}$, there are $\binom{7}{2} = 21$ families. All 24 families are verified computationally in `OctonionicFermat.lean`.

4. General odd primes

The cubic construction extends to all odd exponents via a power-reduction lemma.

Theorem 4.1 (Power Reduction)**.** *Let R be a ring, $u \in R$ with $u^2 = -1$. Then for all $n \geq 0$:*

$$(1 + u)^{2n+1} = (2u)^n(1 + u).$$

Proof. Induction on n. The base case $n = 0$ is immediate. For the inductive step, write $2n + 3 = (2n + 1) + 2$ and compute:

$$\begin{aligned}
(1 + u)^{2n+3} &= (1 + u)^{2n+1} \cdot (1 + u)^2 \\
&= (2u)^n(1 + u) \cdot 2u && \text{(IH and Lemma 2.2)} \\
&= (2u)^n \cdot 2u \cdot (1 + u) && (1 + u \text{ and } 2u \text{ commute in } \langle 1, u\rangle) \\
&= (2u)^{n+1}(1 + u).
\end{aligned}$$

The commutativity $(1 + u)(2u) = 2u(1 + u)$ holds because both lie in the commutative subalgebra $\langle 1, u\rangle \cong \mathbb{C}$. Explicitly: $(1 + u)(2u) = 2u + 2u^2 = 2u - 2$ and $2u(1 + u) = 2u + 2u^2 = 2u - 2$. □

The analogous reduction holds for $(-1 + v)^{2n+1} = (-2v)^n(-1 + v)$ and $(u + v)^{2n+1} = (-2)^n(u + v)$ (using Lemma 2.3).

Theorem 4.2 (All Odd Primes)**.** *Let R be a ring with $u, v \in R$ satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$. Then for every odd $p \geq 3$:*

(i) If $p = 4m + 1$: $(1 + u)^p + (-1 + v)^p = (-1)^m(u + v)^p$.
(ii) If $p = 4m + 3$: $(1 + u)^p + (-1 + v)^p = (-1)^{m+1}(u + v)^p$.

In particular, $a = 1 + u$, $b = -1 + v$, $c = \varepsilon(u + v)$ with $\varepsilon^2 = 1$ satisfies $a^p + b^p = \varepsilon \cdot c^p$.

Proof. We treat the two cases separately.

Case $p = 4m + 1$. By Theorem 4.1 and its analogues:

$$(1 + u)^{4m+1} = (2u)^{2m}(1 + u),$$
$$(-1 + v)^{4m+1} = (-2v)^{2m}(-1 + v).$$

Now $(2u)^{2m} = (4u^2)^m = (-4)^m$ and $(-2v)^{2m} = (4v^2)^m = (-4)^m$. The sum becomes:

$$(-4)^m(1 + u) + (-4)^m(-1 + v) = (-4)^m(u + v).$$

For the right-hand side, $(u + v)^{4m+1} = (-2)^{2m}(u + v) = 4^m(u + v)$, so $(-1)^m(u + v)^{4m+1} = (-1)^m \cdot 4^m(u + v) = (-4)^m(u + v)$.

Case $p = 4m + 3$. By Theorem 4.1:

$$(1 + u)^{4m+3} = (2u)^{2m+1}(1 + u),$$
$$(-1 + v)^{4m+3} = (-2v)^{2m+1}(-1 + v).$$

Now $(2u)^{2m+1} = (-4)^m \cdot 2u$ and $(-2v)^{2m+1} = (-4)^m \cdot (-2v)$. The sum is:

$$(-4)^m\big[2u(1 + u) + (-2v)(-1 + v)\big].$$

Expanding: $2u(1 + u) = 2u + 2u^2 = 2u - 2$ and $(-2v)(-1 + v) = 2v - 2v^2 = 2v + 2$. So the bracket equals $2(u + v)$. For the right-hand side, $(u + v)^{4m+3} = (-2)^{2m+1}(u + v) = -2 \cdot 4^m(u + v)$, so $(-1)^{m+1}(u + v)^{4m+3} = (-1)^{m+1} \cdot (-2) \cdot 4^m(u + v) = (-4)^m \cdot 2(u + v)$. □

Remark 4.3. The sign ε cycles with period 8 in p: $p \equiv 1, 3, 5, 7 \pmod 8$ gives $\varepsilon = +1, -1, -1, +1$. This sign is harmless: since p is odd, $(-c)^p = -(c^p)$, so $a^p + b^p = \varepsilon \cdot c^p$ is equivalent to $a^p + b^p = (\varepsilon c)^p$.

Remark 4.4. The proof is entirely kernel-checked in Lean 4 without `native_decide` or `sorry`: see `AllOddPrimes.lean`, theorems `fermat_all_p_mod1` and `fermat_all_p_mod3`.

5. The dimension boundary

Theorem 5.1 (Dimension Threshold)**.** *Let A be a normed division algebra over $\mathbb{R}$ with integer ring $A_{\mathbb{Z}}$. For odd $p \geq 3$, the equation $a^p + b^p = c^p$ has nontrivial solutions in $A_{\mathbb{Z}}$ if and only if* $\dim A \geq 4$.

Proof. By Hurwitz's theorem (Theorem 2.1), $A \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$. We treat each case.

$\mathbb{H}$ (dim 4) and $\mathbb{O}$ (dim 8): solutions exist. Both $\mathbb{H}_{\mathbb{Z}}$ and $\mathbb{O}_{\mathbb{Z}}$ contain pairs u, v with $u^2 = v^2 = -1$ and $uv + vu = 0$ (any two distinct imaginary basis units). Theorem 4.2 produces solutions for all odd $p \geq 3$. Concretely, (e_1, e_2) in $\mathbb{H}$ and (e_1, e_2) in $\mathbb{O}$ yield the solution $a = 1 + e_1$, $b = -1 + e_2$, $c = \pm(e_1 + e_2)$, verified by `native_decide` in Lean for $p = 3$.

$\mathbb{R}$ (dim 1): FLT holds. This is Wiles' theorem [27]: $a^p + b^p \neq c^p$ for $a, b, c \in \mathbb{Z} \setminus \{0\}$ and $p \geq 3$.

$\mathbb{C}$ (dim 2): FLT holds over $\mathbb{Z}[i]$. The complex numbers have a single imaginary direction. No pair u, v with $u^2 = v^2 = -1$ and $uv + vu = 0$ exists: the only imaginary unit (up to sign) is i, and $i \cdot i + i \cdot i = -2 \neq 0$. The anti-commutativity mechanism does not activate.

Ward [26] established that FLT holds for cubes over $\mathbb{Z}[i]$. More generally, $\mathbb{Z}[i]$ is a commutative PID, so descent arguments (Kummer, Wieferich) apply. Our Lean formalization verifies computationally that no solution exists in $\mathbb{Z}[i]$ with coordinates in $\{-2, \ldots, 2\}$ for $p = 3$ (`DimensionThreshold.lean`, theorem `gaussian_integers_no_cubic_fermat_small`). □

Remark 5.2. The threshold is equivalently characterized: among normed division algebras, $\dim A \geq 4$ if and only if A is non-commutative. The mechanism that breaks FLT is non-commutativity (specifically, anti-commutativity of imaginary units), not non-associativity. The quaternions are associative and already suffice.

Remark 5.3 (Artin confinement). Every known solution to $a^p + b^p = c^p$ in $\mathbb{O}_\mathbb{Z}$ lies in an associative subalgebra $\mathrm{span}\{1, e_i, e_j, e_i e_j\} \cong \mathbb{H}$. Computational evidence (over 600,000 solutions checked) supports the conjecture that all octonionic solutions are confined to quaternionic subalgebras (Conjecture 11.1). If true, the "boundary" for non-commutative Fermat is dimension 4, not 8.

6. Three mechanisms

FLT fails outside $\mathbb{Z}$ through three independent mechanisms.

Theorem 6.1 (Classification of FLT-breaking mechanisms)**.** *Let R be a (possibly non-commutative, possibly non-associative) ring. The equation $a^n + b^n = c^n$ can have nontrivial solutions through:*

- *(A)* ***Anti-commutativity.*** *If R contains u, v with $u^2 = v^2 = -1$, $uv + vu = 0$, then solutions exist for all odd $n \geq 3$.*
- *(B)* ***Zero divisors.*** *If R contains nonzero a, b with $ab = ba = 0$, then $a^n + b^n = (a + b)^n$ for all $n \geq 1$.*
- *(C)* ***Conjugate pairs.*** *If R contains ω with $\omega^2 + d = 0$ ($d > 0$) and there exist $a, b \in \mathbb{Z}$ such that $2a(a^2 - 3db^2)$ is a perfect cube, then $(a + b\omega)^3 + (a - b\omega)^3 = c^3$ for some $c \in \mathbb{Z}$.*

Proof. Mechanism (A) is Theorem 4.2. Mechanism (B) is Theorem 7.1 below. Mechanism (C) is a direct computation:

$$(a + b\omega)^3 + (a - b\omega)^3 = 2a^3 + 6ab^2\omega^2 = 2a^3 - 6ab^2 d = 2a(a^2 - 3db^2).$$

If this equals c^3, we have a Fermat triple. □

Example 6.2. Mechanism (C) produces solutions in commutative rings:

- $\mathbb{Z}[\sqrt{-5}]$: $(-4 - \sqrt{-5})^3 + (-4 + \sqrt{-5})^3 = (-2)^3$. Here $a = -4$, $b = 1$, $d = 5$, and $2(-4)(16 - 15) = -8 = (-2)^3$.
- $\mathbb{Z}[\sqrt{-2}]$: $(-4 - 4\sqrt{-2})^3 + (-4\sqrt{-2})^3 = (-4 + 4\sqrt{-2})^3$.

These counterexamples disprove the original conjecture that anti-commutativity is necessary for FLT to fail. A corrected characterization requires a third disjunct for commutative extensions where the norm form permits cube-sum cancellation.

Remark 6.3. Mechanism (A) is the sharpest, producing solutions where the entire sum lies in the imaginary subspace (real parts cancel to zero). Mechanism (C) accounts for approximately 70% of all solutions found computationally but is harder to prove symbolically for general p. Mechanism (B) is the bluntest: in the sedenions, it eliminates the even/odd distinction entirely.

7. Beyond division: the sedenion phase transition

The Cayley-Dickson doubling continues past the octonions: $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$ are the *sedenions* (dimension 16). They are power-associative and flexible but no longer alternative, and crucially, they contain *zero divisors*: nonzero elements whose product is zero.

Theorem 7.1 (Zero-Divisor Binomial Collapse)**.** *Let R be an associative ring with $a, b \in R$ satisfying $ab = ba = 0$, $a \neq 0$, $b \neq 0$. Then for all $n \geq 1$:*

$$(a + b)^n = a^n + b^n.$$

Setting $c = a + b$ gives $a^n + b^n = c^n$ for all n.

Proof. By induction on n. The base case $n = 1$ is immediate. For $n \geq 1$:

$$\begin{aligned}(a + b)^{n+1} &= (a^n + b^n)(a + b) && \text{(inductive hypothesis)}\\ &= a^{n+1} + a^n b + b^n a + b^{n+1}.\end{aligned}$$

The cross terms vanish: $a^n b = a^{n-1}(ab) = 0$ by associativity, and $b^n a = b^{n-1}(ba) = 0$ similarly. □

Example 7.2 (Sedenion zero-divisor pair)**.** The classic zero-divisor pair of Moreno [21]:

$$\alpha = (e_3, e_2) = e_3 + e_{10}, \qquad \beta = (e_6, -e_5) = e_6 - e_{13}$$

in the Cayley-Dickson representation $\mathbb{S} = \mathbb{O} \times \mathbb{O}$. Both products $\alpha\beta$ and $\beta\alpha$ vanish, verified by `native_decide` on the 16-dimensional integer computation (`SedenionZeroDivisor.lean`). Theorem 7.1 then gives $\alpha^n + \beta^n = (\alpha + \beta)^n$ for all $n \geq 1$.

Remark 7.3. This is a *phase transition* in the Cayley-Dickson tower. In the quaternions and octonions (division algebras), FLT breaks for odd exponents but holds for even ones: if $p = 2m$ with $m \geq 3$, then a^{2m} is a real scalar (equal to $\pm\mathrm{N}(a)^m$), so the equation reduces to $\pm\mathrm{N}(a)^m \pm \mathrm{N}(b)^m = \pm\mathrm{N}(c)^m$, which is classical FLT. In the sedenions, zero divisors make norms non-multiplicative, and the even/odd distinction vanishes.

More precisely, the Cayley-Dickson algebra at dimension 2^k has $\binom{2^{k-1}-1}{2}$ anti-commuting pair families (Mechanism A), which grow as $O(4^k)$. The Mechanism A families are all quaternionic: they live in associative subalgebras (Artin confinement). Mechanism B families appear only for $k \geq 4$ (dimension ≥ 16) but then apply to all exponents.

8. Solution classification and computational data

8.1. G_2 **orbit collapse.** The 21 octonionic solution families are indexed by pairs (e_i, e_j) with $1 \leq i < j \leq 7$. The automorphism group of the octonions is the exceptional Lie group $G_2 = \mathrm{Aut}(\mathbb{O})$, a 14-dimensional compact Lie group.

Theorem 8.1 (G_2 Orbit Collapse)**.** *All 21 solution families form a single orbit under G_2.*

Proof. Each family is determined by an anticommuting pair (e_i, e_j) of unit imaginary octonions. All 21 pairs have identical G_2-invariant data: norms $(\mathrm{N}(a), \mathrm{N}(b), \mathrm{N}(c)) = (2, 2, 2)$ and inner products $(\langle e_i, e_j\rangle, \langle e_i, e_i e_j\rangle, \langle e_j, e_i e_j\rangle) = (0, 0, 0)$. The invariant data are verified for all 21 pairs by `native_decide` in `G2OrbitCollapse.lean`.

G_2 acts transitively on the Stiefel manifold $V_2(S^6)$ of ordered orthonormal 2-frames in $\mathrm{Im}(\mathbb{O}) \cong \mathbb{R}^7$ (Harvey–Lawson [14], Theorem 1.38). Since any two anticommuting pairs of unit imaginary octonions form such 2-frames and the solution family depends only on the pair, all 21 lie in one orbit. □

Table 1. Solution counts by algebra and exponent

Algebra	Dim	Families (Mech. A)	Even p?	All n?
$\mathbb{R}$	1	0	No	No
$\mathbb{C}$	2	0	No	No
$\mathbb{H}$	4	$\binom{3}{2} = 3$	No	No
$\mathbb{O}$	8	$\binom{7}{2} = 21$	No	No
$\mathbb{S}$	16	$\binom{15}{2} = 105$	Yes (Mech. B)	Yes

8.2. **Computational data.**

8.3. **Waring bounds.** In $\mathbb{Z}$, the Waring number $g(3) = 9$: every integer is a sum of at most 9 cubes. The quaternionic integers appear to have much smaller Waring numbers.

Computational data for all 4,785 Lipschitz integers with $\mathrm{N}(q) \leq 30$:

- $g(3, \mathbb{H}_\mathbb{Z}) \leq 4$. Only 12 elements (all at norm 26) require 4 cubes.
- $g(5, \mathbb{H}_\mathbb{Z}) \leq 5$, $g(7, \mathbb{H}_\mathbb{Z}) \leq 6$.

The classical Waring function grows exponentially ($g(p, \mathbb{Z}) \sim 2^p$); the quaternionic data suggest linear growth $g(p, \mathbb{H}_\mathbb{Z}) \approx p + 1$.

9. The Langlands bridge

The four normed division algebras organize a stratification of the Langlands program.

Table 2. The division algebra stratification of the Langlands program

Level	Algebra	Lost	Group	Status
1	$\mathbb{R}$ (dim 1)	—	GL_1	Complete (Artin, 1927)
2	$\mathbb{C}$ (dim 2)	ordering	GL_2	Largely complete (Wiles, 1995)
3	$\mathbb{H}$ (dim 4)	commutativity	GL_2 inner forms	Complete (JL, 1970)
4	$\mathbb{O}$ (dim 8)	associativity	G_2	Local complete, global open

9.1. **Four levels.** Fermat's Last Theorem lives at Level 2. The proof passes through Level 3: Ribet's level-lowering uses the Jacquet-Langlands correspondence (inner forms of GL_2 over quaternion algebras) to transfer modular representations between Shimura curves and modular curves. The octonionic extension (this paper) probes Level 4, where the automorphism group $G_2 = \mathrm{Aut}(\mathbb{O})$ replaces GL_2.

9.2. **The Shimura obstruction.**

Theorem 9.1 (Shimura Obstruction at Level 4)**.** *The symmetric space $G_2(\mathbb{R})/\mathrm{SO}(4)$ is not Hermitian symmetric. Consequently, G_2 does not admit a Shimura variety.*

Proof. A connected reductive group G over $\mathbb{R}$ admits a Shimura variety if and only if its symmetric space $G(\mathbb{R})/K$ is Hermitian symmetric, i.e., admits a $G(\mathbb{R})$-invariant complex structure. For G_2, the maximal compact subgroup is $K \cong \mathrm{SO}(4)$ and $\dim(G_2(\mathbb{R})/\mathrm{SO}(4)) = 14 - 6 = 8$, which is even. However, G_2 is an exceptional group of rank 2, and its restricted root system is of type G_2 (not type BC_1). A Hermitian symmetric space of non-compact type has restricted root system of type C_r or BC_r; since $G_2 \neq C_r$ for any r, the space is not Hermitian. This is formalized in `Bridge.lean`, theorem `no_shimura_at_octonionic_level`. □

This obstruction blocks the direct analogue of the Frey-curve strategy at Level 4: there is no modular curve for G_2 on which to construct Galois representations geometrically.

9.3. **The Gross-Savin program.** The viable route to G_2 Galois representations passes through the Gross-Savin program [13].

The idea: automorphic forms on the compact form of G_2 theta-lift to Siegel cusp forms on GSp_6. The Siegel modular variety for GSp_6 *is* a

Shimura variety, so Galois representations exist by the construction of Kret–Shin [17]. The Satake parameter constraint forces the image to lie in $G_2(\mathbb{Q}_\ell) \subset \mathrm{GSp}_6(\mathbb{Q}_\ell)$.

The program has advanced substantially:

- Steps 1–2 (theta lift to nonzero Siegel cusp form): proved by Gross–Savin [13].
- Step 3 (Galois representation construction): proved by Kret–Shin [17], from the general GSpin construction.
- Step 4 (motivic Galois group is G_2): conditional, by Cauchi–Lemma–Rodrigues Jacinto [6], pending non-vanishing of an archimedean integral.

Global modularity lifting ($R = T$ for G_2) remains wide open.

9.4. **Connection to Bhargava's higher composition laws.** The split form of G_2 acts on $\mathbb{Z}^2 \otimes \mathbb{Z}^3 \otimes \mathbb{Z}^3$ (Zorn's vector matrices), parametrizing cubic rings. Bhargava [2] showed that $\mathrm{GL}_2(\mathbb{Z}) \times \mathrm{GL}_3(\mathbb{Z})$-orbits on $2 \times 3 \times 3$ cubes of integers parametrize pairs of ideal classes in cubic rings.

Our construction connects: the octonionic Fermat solutions live on the trilinear variety $V(\varphi) = \{(a, b, c) \in (\mathbb{Z}^7)^3 : \varphi(a, b, c) = 0\}$ where $\varphi(a, b, c) = \langle c, a \times b\rangle$ is the G_2-invariant trilinear form and $\times$ is the octonionic cross product. The $G_2(\mathbb{Z})$-orbits on Zorn integer matrices yield cubic forms; computational data (6,560 matrices $\to$ 9,565 cubic forms $\to$ 47,961 cubic rings, all with discriminant $\equiv 0 \pmod 4$) suggest a parametrization theorem connecting the octonionic integer lattice to Bhargava's program.

9.5. **The magic square.** The Freudenthal-Tits magic square [10] organizes the relationship between division algebras and Lie algebras. Given two normed division algebras A, B, the construction $M(A, B) = \mathrm{Der}(A) \oplus (A_0 \otimes J_0) \oplus \mathrm{Der}(J)$ (where $J = J_3(B)$) produces:

	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	$\mathbb{O}$
$\mathbb{R}$	A_1	A_2	C_3	F_4
$\mathbb{C}$	A_2	$A_2 \oplus A_2$	A_5	E_6
$\mathbb{H}$	C_3	A_5	D_6	E_7
$\mathbb{O}$	F_4	E_6	E_7	E_8

The octonionic column produces all five exceptional Lie algebras (G_2, F_4, E_6, E_7, E_8). The entry $M(\mathbb{O}, \mathbb{O}) = E_8$ connects the octonionic Fermat program to the E_8 lattice (Viazovska [25]) and, speculatively, to the Langlands program for exceptional groups.

10. Machine verification

All theorems in this paper are formalized in Lean 4 with the Mathlib library. The formalization is structured as follows:
Totals: 28 files, 8,791 lines, 355 theorems and lemmas.

The formalization uses two proof strategies:

TABLE 3. Key files in the Lean formalization

File	Content	Key Theorems
`Algebra.lean`	Octonion/quaternion integer definitions	—
`OctonionicFermat.lean`	Cubic verification (`native_decide`)	Thm. 3.1
`SymbolicFermatGeneral.lean`	Power reduction (kernel-checked)	Thm. 4.1
`AllOddPrimes.lean`	All odd p (kernel-checked)	Thm. 4.2
`DimensionThreshold.lean`	Dimension boundary	Thm. 5.1
`ZeroDivisorFermat.lean`	Zero-divisor collapse (kernel-checked)	Thm. 7.1
`SedenionZeroDivisor.lean`	Sedenion instantiation	Ex. 7.2
`G2OrbitCollapse.lean`	G_2 orbit analysis	Thm. 8.1
`Bridge.lean`	Langlands stratification	Thm. 9.1
`Conjectures.lean`	17 conjectures (documentation)	—

(1) **Kernel-checked symbolic proofs** for the main theorems (Theorems 4.1, 4.2, 7.1). These use only ring axioms and structural induction, with no `native_decide` and no `sorry`. They are valid in any ring satisfying the hypotheses.
(2) **Computational verification** (`native_decide`) for concrete instances: specific triples in $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$, all 21 family verifications, sedenion zero-divisor products, and the Gaussian integer exhaustive search. These verify that the abstract hypotheses are satisfied by the concrete algebras.

The entire formalization builds without error, `sorry`, or axiom declarations beyond Lean's foundational axioms (`propext`, `Quot.sound`, `Classical.choice`).

11. Open problems and conjectures

The octonionic Fermat program generates a constellation of precise, falsifiable conjectures. We state ten, selected from a registry of 17 (the full list is documented in `Conjectures.lean`).

Conjecture 11.1 (Artin Confinement — A4)**.** Every solution to $a^p+b^p = c^p$ in $\mathbb{O}_\mathbb{Z}$ (p odd, ≥ 3) lies in an associative subalgebra $\cong \mathbb{H}$.

Evidence. Over 600,000 solutions checked; zero have nonzero associator. The orbit analysis found 61,488 solutions at $\mathrm{N}() \leq 8$, all with $[a, b, c] = 0$. *Expected* by Artin's theorem: any two elements of an alternative algebra generate an associative subalgebra.

Conjecture 11.2 (Waring Tightness — B1)**.** $g(3, \mathbb{H}_\mathbb{Z}) = 4$.

Evidence. Verified for all 4,785 Lipschitz integers with $\mathrm{N}(q) \leq 30$. Exactly 12 elements (all at norm 26) require 4 cubes.

Conjecture 11.3 (Split G_2 Parametrization — D1)**.** Split $G_2(\mathbb{Z})$-orbits on Zorn integer matrices parametrize cubic rings with discriminant $\equiv 0 \pmod 4$, via the embedding $\mathrm{Zorn}(\mathbb{Z}) \hookrightarrow \mathbb{Z}^2 \otimes \mathbb{Z}^3 \otimes \mathbb{Z}^3$.

Evidence. 6,560 Zorn matrices yield 9,565 cubic forms and 47,961 cubic rings, all with discriminant $\equiv 0 \pmod 4$, all associative.

Conjecture 11.4 (G_2 Fourier Coefficients and Fermat Families — E4)**.** The theta lift of automorphic forms on G_2^c to GSp_6 produces Siegel modular forms whose Fourier coefficients encode the number of octonionic Fermat solution families at each norm level.

Conjecture 11.5 (G_2 Selmer Averaging — E2)**.** $\operatorname{avg} |\mathrm{Sel}_2(G_2, V)| = 3$, by analogy with the Bhargava-Shankar theorem for elliptic curves.

Status. Speculative. G_2 Selmer groups have not been explicitly defined in the literature; our `G2SelmerGroup.lean` is a first attempt.

Conjecture 11.6 (Hasse Principle for $V(\varphi)$ — C2)**.** For the primitive subfamily $V(\varphi)_{\mathrm{prim}}$ (where $a \times b \neq 0$ and a, b, c are pairwise non-proportional), the Hasse principle holds: $V(\varphi)_{\mathrm{prim}}(\mathbb{Q}) \neq \varnothing$ if and only if $V(\varphi)_{\mathrm{prim}}(\mathbb{Q}_v) \neq \varnothing$ for all places v.

Evidence. No local-global failures found; the octonionic cross product is surjective modulo every prime tested ($p \leq 13$), so no Brauer-Manin obstruction at finite primes.

Conjecture 11.7 (No Third Mechanism — S3)**.** Every nontrivial solution to $a^n + b^n = c^n$ in $\mathbb{S}_{\mathbb{Z}}$ (for $n \geq 3$) arises from either Mechanism A (anti-commutativity) or Mechanism B (zero divisors).

Conjecture 11.8 (Waring Linear Growth — B2)**.** $g(p, \mathbb{H}_{\mathbb{Z}}) \sim p + 1$ for odd primes p.

Evidence. Three data points: $g(3) \leq 4$, $g(5) \leq 5$, $g(7) \leq 6$.

Conjecture 11.9 (Zariski Density — C3)**.** $V(\varphi)(\mathbb{Q})$ is Zariski-dense in $V(\varphi)$.

Argument. The fiber structure $\varphi(a, b, c) = \langle c, a \times b \rangle$ gives codimension-1 linear fibers in c for each (a, b) with $a \times b \neq 0$. Dense base and dense fibers imply dense total space.

Conjecture 11.10 (φ-Contraction as G_2 Curvature — F3)**.** The symmetrized contraction identity for the octonionic trilinear form φ (proved algebraically in `TrilinearVariety.lean`) encodes the curvature of G_2-holonomy manifolds in the sense of Bryant [4] and Joyce [16].

References

[1] J. C. Baez, *The octonions*, Bull. Amer. Math. Soc. **39** (2002), 145–205.
[2] M. Bhargava, *Higher composition laws I: A new view on Gauss composition, and quadratic generalizations*, Ann. Math. **159** (2004), 217–250.
[3] M. Bhargava and A. Shankar, *Binary quartic forms having bounded invariants, and the boundedness of the average rank of elliptic curves*, Ann. Math. **181** (2015), 191–242.

[4] R. L. Bryant, *Metrics with exceptional holonomy*, Ann. Math. **126** (1987), 525–576.
[5] A. Ben-Zvi, Y. Sakellaridis, and A. Venkatesh, *Relative Langlands duality*, preprint (2024).
[6] A. Cauchi, F. Lemma, and J. Rodrigues Jacinto, *Algebraic cycles and functorial lifts from G_2 to* PGSp_6, Algebra Number Theory (2025).
[7] J. H. Conway and D. A. Smith, *On Quaternions and Octonions*, A. K. Peters, 2003.
[8] N. Elkies and B. Gross, *Cubic rings and the exceptional Jordan algebra*, Duke Math. J. **109** (2001), 383–409.
[9] L. Fargues and P. Scholze, *Geometrization of the local Langlands correspondence*, preprint (2024).
[10] H. Freudenthal, *Beziehungen der E_7 und E_8 zur Oktavenebene, I–XI*, Indagationes Math. (1954–1964); J. Tits, *Algèbres alternatives, algèbres de Jordan et algèbres de Lie exceptionnelles*, Indag. Math. **28** (1966), 223–237.
[11] D. Gaitsgory et al., *Proof of the geometric Langlands conjecture*, preprint (2024).
[12] W. T. Gan, B. Gross, and G. Savin, *Fourier coefficients of modular forms on G_2*, Duke Math. J. **115** (2002), 105–169.
[13] B. Gross and G. Savin, *Motives with Galois group of type G_2: an exceptional theta-correspondence*, Compositio Math. **114** (1998), 153–217.
[14] R. Harvey and H. B. Lawson, *Calibrated geometries*, Acta Math. **148** (1982), 47–157.
[15] A. Hurwitz, *Über die Composition der quadratischen Formen von beliebig vielen Variablen*, Nachr. Ges. Wiss. Göttingen (1898), 309–316.
[16] D. Joyce, *Compact Manifolds with Special Holonomy*, Oxford University Press, 2000.
[17] A. Kret and S. W. Shin, *Galois representations for general symplectic groups*, J. Eur. Math. Soc. (2023).
[18] S. Krutelevich, *Jordan algebras, exceptional groups, and Bhargava composition*, J. Algebra **314** (2007), 924–977.
[19] S. Leslie and A. Pollack, *Quaternionic modular forms of degree two*, preprint (2024).
[20] J. Li and Y. Yuan, *Fermat's equation over 2×2 integer matrices*, preprint (2025).
[21] G. Moreno, *The zero divisors of the Cayley-Dickson algebras over the real numbers*, Bol. Soc. Mat. Mexicana **4** (1998), 13–28.
[22] A. Pollack, *The minimal modular form on the exceptional group G_2*, preprint (2018).
[23] S. Reggiani, *On the structure of Cayley-Dickson algebras*, preprint (2024).
[24] P. Ribenboim, *My Numbers, My Friends: Popular Lectures on Number Theory*, Springer, 2000; *Fermat's Last Theorem for Amateurs*, Springer, 1999.
[25] M. Viazovska, *The sphere packing problem in dimension 8*, Ann. Math. **185** (2017), 991–1015.
[26] M. Ward, *Fermat's Last Theorem for Gaussian integers*, unpublished note (2008).
[27] A. Wiles, *Modular elliptic curves and Fermat's Last Theorem*, Ann. Math. **141** (1995), 443–551.

7331
Email address: `tim@7331.ink`